Vinay Kumar Mandadi

Academic CV

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EDUCATION

Lovely Professional University (LPU), Punjab, India Aug 2022 – Jul 2026
B.Tech, Computer Science and Engineering (specialisation: Google Cloud Computing)

Capstone: Truth in Pixels: Advancing Deep Fake Models for Reliable Deepfake Detection Jan – May 2026
Team of five, supervised by Harjit Kaur. EfficientNetV2-M, BiLSTM and attention fused with a DCT frequency branch; AUC-ROC 0.968 to 0.992 on FaceForensics++ (C23), 91.4% to 93.1% accuracy on Celeb-DF v2 (cross-dataset); Grad-CAM, ONNX export, Flask app.

RESEARCH INTERESTS

Software engineering for reliable systems: CI/CD failure prediction, software risk prediction, and trustworthy, calibrated machine learning in operations (AIOps). AI systems: multi-agent validation of LLM outputs.

PUBLICATIONS AND PREPRINTS

Explainable and Trustworthy Network Intrusion Detection: From Multi-Class SHAP Analysis to Confidence-Aware Deployment
Sole author. Accepted at IEEE IS'26, 13th International Conference on Intelligent Systems; not presented. Preprint: doi.org/10.5281/zenodo.21260892.
99.02% accuracy (macro F1 0.906) on CIC-IDS2017, collapsing to 31.94% with 0% attack recall on UNSW-NB15 without retraining. Confidence filtering fails under this shift; Platt scaling restores 68.0% out-of-domain accuracy (ECE 0.622 to 0.097).

AIRIMF: An AI-Driven Risk Identification and Mitigation Framework for Software Project Management
Sole author. Preprint: doi.org/10.5281/zenodo.20691021.
Predicts high-risk issues the moment a ticket is filed: 76.21% cross-validated accuracy and 82% high-risk recall on VS Code (2,593 bug reports), 83.37% and 89% on React (744); significantly better than SVM (McNemar, $p < 0.0001$).

Combating Synthetic Media Manipulation Using Deep Learning-Based Deepfake Detection Systems
Co-author (five LPU students with supervisor Harjit Kaur). Accepted at 8BIC 2026.
Proposes AFDDA, which combines spatial CNN features with temporal frame-difference features into a weighted deepfake artifact score.

RESEARCH AND TECHNICAL PROJECTS

DevSignal HQ CI failure prediction · 2026
Python · Flask · SQLite · Resend · Railway · GitHub Apps

- GitHub App that scores the failure risk of a CI workflow before it runs and emails an alert; JWT app authentication and automatic webhook setup on install.
- Built and deployed solo on Railway, based on research into GitHub Actions failure prediction.

Veridian and AMAV Multi-agent LLM validation
Python · FastAPI · Gemini · GPT-4o-mini · Tavily · MCP · Kubernetes · github.com/vinaykumarchowdary18/veridian

- Veridian (Kaggle AI Agents Intensive Capstone 2026, Agents for Business): an analyst model drafts an answer from live retrieved evidence, two auditors from different model families critique it, and an arbiter outputs a consensus score and a return, revise or escalate decision.
- Treats retrieved web content as untrusted: prompt-injection screening, input bounds, rate limiting, audit logging. Predecessor AMAV runs containerised on Kubernetes.

ClickFix First Aid Open source (MIT) · Oct 2026
JavaScript · Chrome/Edge Manifest V3 · PowerShell · GitHub Actions · Playwright · github.com/vinaykumarchowdary18/clickfix-first-aid

- Browser extension that stops fake "I'm not a robot" (ClickFix) pages from putting malware commands on the clipboard, plus a read-only Windows check and a recovery guide.
- 248 automated tests (engine, PowerShell and real-browser end-to-end); CI on Linux and Windows with a Microsoft Defender scan; reproducible releases with SHA-256 checksums.
- Submitted to Microsoft Edge Add-ons (in review).

PaperLens AI-text detection and plagiarism SaaS

FastAPI · React · Vite · SQLite · Google OAuth · Cashfree · github.com/vinaykumarchowdary18/PaperLens

- Solo full-stack build. FastAPI backend runs a 7-agent AI-text detection ensemble and a 6-source plagiarism check (Semantic Scholar, OpenAlex, CrossRef, CORE, Wikipedia, self-check).
- Google OAuth with JWT, Cashfree payments (UPI, cards, net banking) with a credit system, rate limiting and async background jobs; React and Vite front end with live result polling.

EXPERIENCE

AI Evaluation Contributor (freelance), Handshake AI Remote · Jul 2026 – Present

- Write hard, multi-step research questions that current AI models get wrong, with answer trails verified against independent primary sources, used to evaluate and train AI systems. All submitted tasks approved on human review.

TEST SCORES AND LANGUAGES

GRE General Test: 324 (Quantitative 168, Verbal 156, Analytical Writing 3.5), Dec 2025

English (professional; degree taught in English), Telugu (native), Hindi, Japanese (elementary; studying toward JLPT N2)

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Multicloud Architect Professional Oct 2025
- Oracle Cloud Infrastructure 2025 Certified DevOps Professional Oct 2025
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate Oct 2025
- Oracle Data Platform 2025 Certified Foundations Associate May 2026

SKILLS

Languages: Python, TypeScript, JavaScript, Java, Bash, SQL

Backend and web: FastAPI, Flask, Django, Node.js, Express, React, Vite, REST APIs, OAuth and JWT

Cloud and DevOps: Oracle Cloud (OCI), Google Cloud, AWS, Docker, Kubernetes, Terraform, GitHub Actions, Railway, Firebase, Linux

AI and ML: scikit-learn, pandas, NumPy, PyTorch, TensorFlow, SHAP, model calibration, LLM APIs (Gemini, OpenAI), multi-agent pipelines

AWARDS

- GDG Solution Challenge 2025: top 105 of 3,700+ teams, with the Secure Aadhaar Voting Platform
- Google Cloud Skills Boost: Gold tier